

CMAN :=

①

Given $N = 1024$, $f = 50 \text{ Hz}$

Task 1 :=

$$R = N \times f = 1024 \times 50 = 51,200 \text{ spikes/second}$$

Task 2 :=

$B = R \times 20$ (bits per packet total)

$$B = 51,200 \times 20 = 1,024,000 \text{ bits/seconds}$$

$$B = 1.024 \text{ Mbits/s}$$

Task 3 :=

Given, SPI ($\leq 50 \text{ Mbit/s}$), I^2C ($\leq 3.4 \text{ Mbit/s}$),
AXI4-Lite (assume 100 Mbit/s)

from task 2, $B = 1.024 \text{ Mbits/s}$

$$\text{max} \geq B$$

$$I^2C \Rightarrow 3.4 \text{ Mbit/s} \geq 1.024 \text{ Mbit/s}$$

$$\text{SPI} \Rightarrow 50 \text{ Mbit/s} \geq 1.024 \text{ Mbit/s}$$

$$\text{AXI4} \Rightarrow 100 \text{ Mbit/s} \geq 1.024 \text{ Mbit/s}$$

So, I^2C is the lowest complexity than SPI and AXI4-lite.

Task 4 :=

25% of 1024 neurons fire at once

$$\frac{25 \times 1024}{100} = 0.25 \times 1024$$

Neuron firing = 256 Neurons

Each neuron that fire sends 1 Packet = 20 bits (From task 2)

$$\begin{aligned}\text{Total bits} &= \text{Neuron firing} \times \text{bits per Packet} \\ &= 256 \times 20 = 5,120 \text{ bits}\end{aligned}$$

$$\text{Total bits} = 5120 \text{ bits.}$$

$$1 \text{ ms} = \frac{1}{1000} \text{ second} = 0.001 \text{ Seconds}$$

$$\text{Peak bandwidth} = \frac{5120}{0.001} = 5120000 \text{ bit/s}$$

$$\text{From task 2, Bandwidth} = 5.12 \text{ Mbit/s}$$

$$B_{\text{mean}} = 1.024 \text{ Mbits/s}$$

$$\text{Burst Bandwidth (} B_{\text{burst}} \text{)} = 5.12 \text{ Mbit/s}$$

$$\text{Ratio} = \frac{B_{\text{burst}}}{B_{\text{mean}}} = \frac{5.12}{1.024} = 5 \times$$

From task 3, we chose I^2C

$$3.4 > 5.12, \text{ NO, it's too slow.}$$

Here we have multiple choices

Choice A: We can Add a buffer.

$$\text{Buffer depth} = \text{Burst Packets} = 256 \text{ Packets}$$

Choice B := Upgrade to SPI

$$50 \geq 5.12, \text{ it handles the burst easily.}$$

No buffer is needed in this SPI.

Task 5: =

Given, All 1024 neurons and sends 1bit Per neuron fired.

$$\begin{aligned}\text{Bits per frame} &= 1024 \text{ neurons} \times 1 \text{ bit per neuron} \\ &= 1024 \text{ bits per frame}\end{aligned}$$

$$\text{Frame per second} = \frac{1 \text{ second}}{1 \text{ ms}} = \frac{1}{0.001} = 1000 \frac{\text{frames}}{\text{second}}$$

$$\begin{aligned}B_{\text{frame}} &= \text{Bits per frame} \times \text{frames per second} \\ &= 1024 \times 1000 = 1024000 \text{ bit/s}\end{aligned}$$

$$B_{\text{frame}} = 1.024 \text{ Mbit/s}$$

$$\text{From task 2: } BAER = 1.024 \text{ Mbit/s}$$

$$\text{Ratio} = \frac{BAER}{B_{\text{frame}}} = \frac{1.024 \text{ Mbit/s}}{1.024 \text{ Mbit/s}}$$

$$\text{Ratio} = 1$$

$$BAER = B_{\text{frame}}$$

$$N \times f \times 20 = N \times 1000 \times 1$$

$$f \times 20 = 1000 \times 1$$

$$f = \frac{1000}{20} = 50$$

$$\boxed{f_{\text{crossover}} = 50 \text{ Hz}}$$

$f < 50\text{Hz}$ $\leftrightarrow$ $f = 50\text{Hz}$ $\leftrightarrow$ $f > 50\text{Hz}$
AER wins equal frame wins

$B_{AER} < B_{frame}$, Because AER uses less Bandwidth because most neurons are quiet, no point sending empty data.

When neurons fire fast ($f > 50\text{Hz}$)

$B_{AER} > B_{frame}$

AER uses more bandwidth because every packet has 20 bits of overhead but frame based only uses 1 bit per neuron.

So, AER is the right choice when your neurons are mostly quiet because why waste bandwidth reporting nothing happened for hundreds of neurons when we can just send a small packet only when something actually fires.